

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories, demographics, and subscription behavior.



Dataset at a Glance

3,900

Total Purchases

18

Features

37

Missing Values

Review Rating column

2

Gender Groups

Male (2,652) / Female

Key features: demographics, purchase details, shopping behavior, subscription status, shipping type, discounts.



Data Preparation in Python

→ Load & Explore

pandas: `df.info()`, `.describe()`

→ Clean & Standardize

Median imputation, snake_case renaming

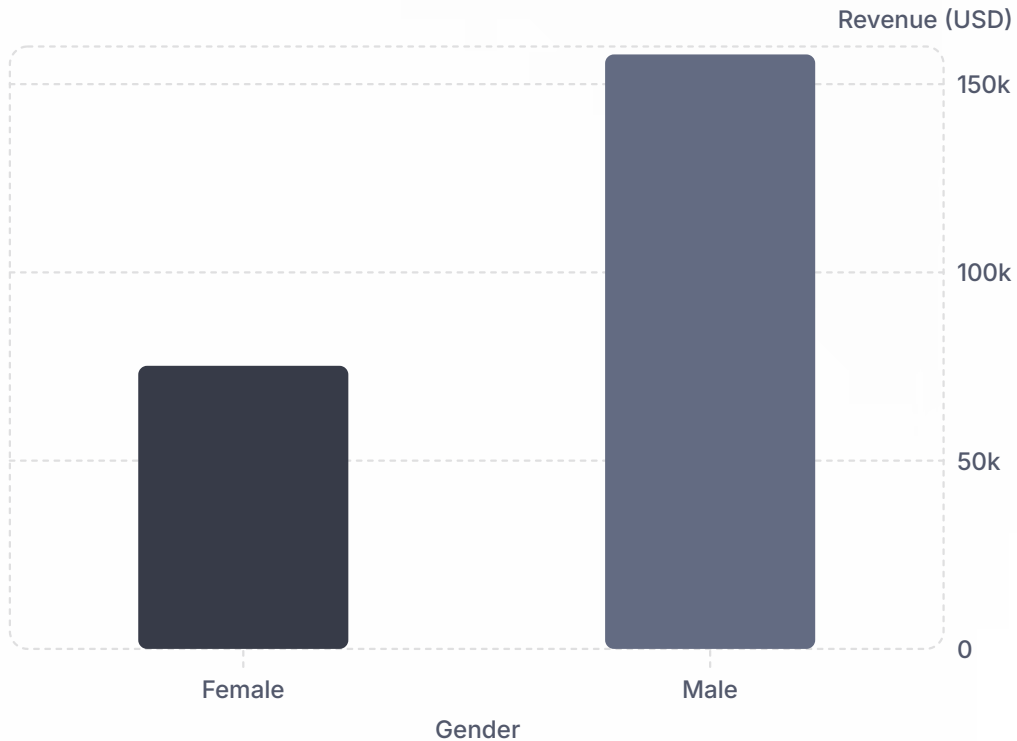
→ Feature Engineering

age_group bins, purchase_frequency_days

→ Load to PostgreSQL

Cleaned DataFrame → SQL analysis

Revenue by Gender

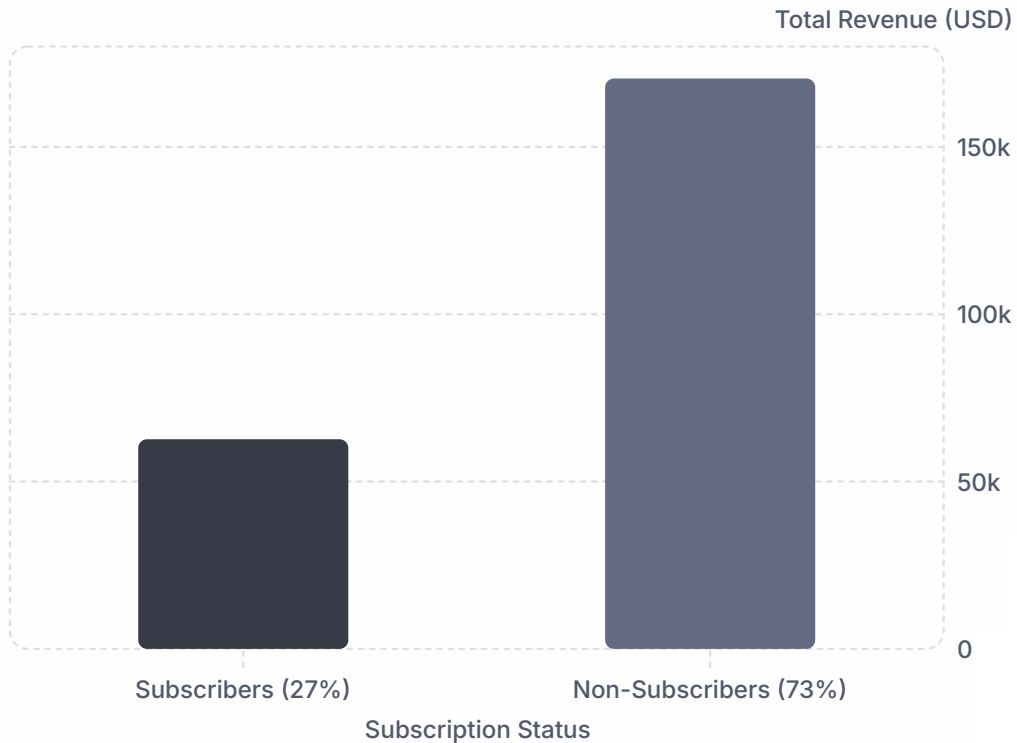


Key Finding

Male customers generated **2.1× more revenue** than female customers — \$157,890 vs \$75,191.

📄 Male: 2,652 customers · Female: 1,248 customers

Subscribers vs. Non-Subscribers



Spending Behavior

Average spend nearly identical: **\$59.49** (subscribers) vs **\$59.87** (non-subscribers).

Non-subscribers drive **73% of total revenue** — opportunity to convert.

Top Products & Discount Patterns

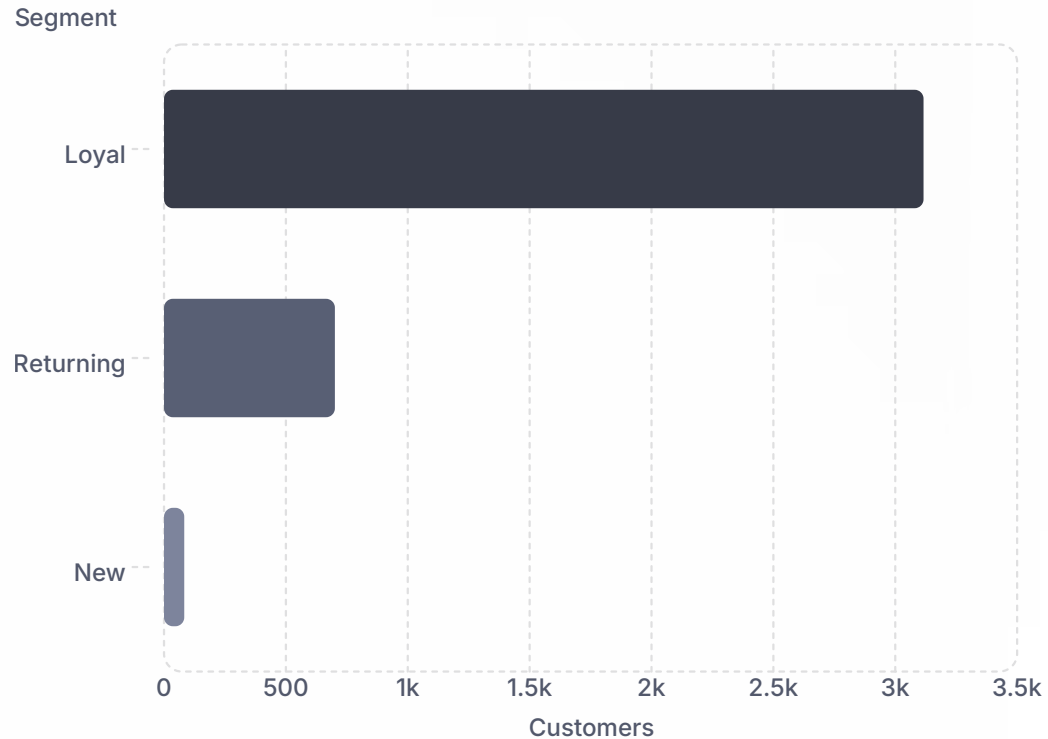
Top 5 by Review Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rate

Product	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation

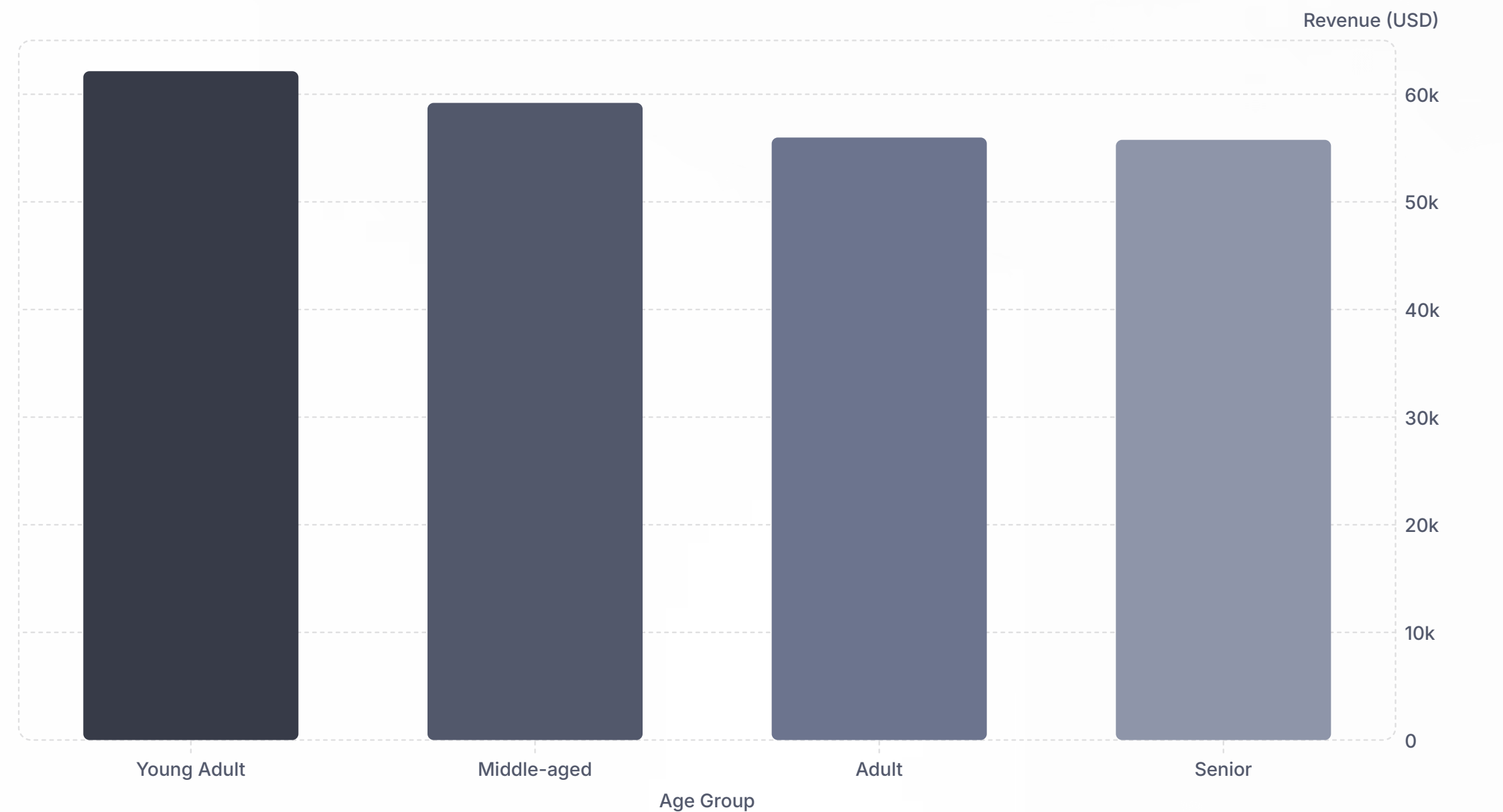


Segment Breakdown

80% of customers are Loyal (3,116). Only 83 New customers — retention is strong, acquisition is the gap.

i Repeat buyers (>5 purchases): 958 subscribers vs 2,518 non-subscribers

Revenue by Age Group



Young Adults lead revenue at **\$62,143**. Revenue is relatively evenly distributed across all age groups — broad targeting is viable.

Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Clothing

Top Category — \$104K revenue, 1,737 sales



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discount Policy

Balance sales boosts with margins



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

